# Chapter 12
# Lookback Options

Lookback call (resp. put) options are financial derivatives that allow their holders to exercise the option by setting the strike price at the minimum (resp. maximum) of the underlying asset price process $(S_t)_{t\in[0,T]}$ over the time interval $[0,T]$. Lookback options can be priced by PDE arguments or by computing the discounted expected values of their claim payoff $C$, namely $C = S_T - \min_{0\leqslant t\leqslant T} S_t$ in the case of call options, and $C = \max_{0\leqslant t\leqslant T} S_t - S_T$ in the case of put options.


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## 12.1 The Lookback Put Option

The standard lookback put option gives its holder the right to sell the underlying asset at its historically highest price. In this case, the floating strike price is $M_0^T$ and the payoff is given by the terminal value

$$C = M_0^T - S_T$$

of the drawdown process $(M_0^t - S_t)_{t\in[0,T]}$. The following pricing formula for lookback put options is a direct consequence of Proposition 10.9, see also Eq. (10) in Goldman et al. (1979).

**Proposition 12.1.** *The price at time $t \in [0,T]$ of the lookback put option with payoff $M_0^T - S_T$ is given by*




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$$
\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[M_0^T - S_T \,\big|\, \mathcal{F}_t\big] \\
&= M_0^t\,\mathrm{e}^{-(T-t)r}\Phi\left(-\delta_-^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) + S_t\left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) \\
&\quad - S_t\,\mathrm{e}^{-(T-t)r}\frac{\sigma^2}{2r}\left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right) - S_t,
\end{aligned}
$$

*where* $\delta_\pm^T(s)$ *is defined in* (11.6).

*Proof.* We have

$$
\begin{aligned}
\mathbb{E}^*\big[M_0^T - S_T \,\big|\, \mathcal{F}_t\big] &= \mathbb{E}^*\big[M_0^T \,\big|\, \mathcal{F}_t\big] - \mathbb{E}^*[S_T \mid \mathcal{F}_t] \\
&= \mathbb{E}^*\big[M_0^T \,\big|\, \mathcal{F}_t\big] - \mathrm{e}^{(T-t)r}S_t,
\end{aligned}
$$

hence Proposition 10.9 shows that

$$
\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[M_0^T - S_T \,\big|\, \mathcal{F}_t\big] \\
&= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[M_0^T \,\big|\, \mathcal{F}_t\big] - \mathrm{e}^{-(T-t)r}\mathbb{E}^*[S_T \mid \mathcal{F}_t] \\
&= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[M_0^T \,\big|\, M_0^t\big] - S_t \\
&= M_0^t\,\mathrm{e}^{-(T-t)r}\Phi\left(-\delta_-^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) - S_t\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) \\
&\quad + S_t\frac{\sigma^2}{2r}\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) - S_t\frac{\sigma^2}{2r}\,\mathrm{e}^{-(T-t)r}\left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right).
\end{aligned}
$$

□

Figure 12.1 represents the lookback put option price as a function of $S_t$ and $M_0^t$, for different values of the time to maturity $T-t$.

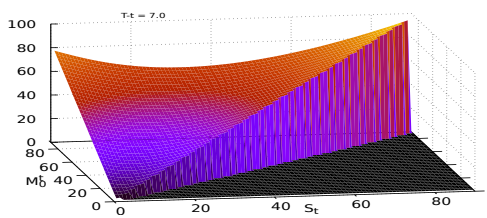
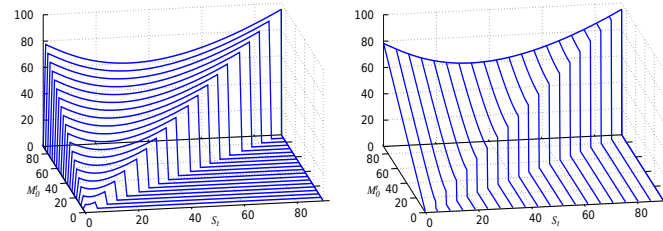



Fig. 12.1: Graph of the lookback put option price (3D).*

From Figures 12.1 and 12.2, we make the following observations.

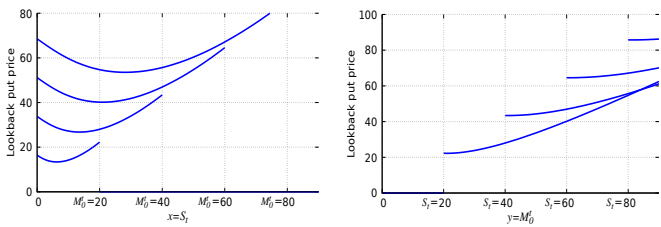
i) Close to maturity, if the underlying asset price $S_t$ is close to $M_0^t$ then an increase in the value $S_t$ can result into a higher put option price, as in this case a variation of $S_t$ may increase the value of $M_0^t$.
ii) When the underlying asset price $S_t$ is far from $M_0^t$, an increase in $S_t$ is less likely to affect the value of $M_0^t$ when time $t$ is close to maturity $T$, and this results into a lower option price.




(a) Put prices as functions of $S$. (b) Put prices as functions of $M$.

Fig. 12.2: Graph of lookback put option prices.

Figures 12.2 and 12.3 show accordingly that, from the Delta hedging strategy for lookback put options, see Proposition 12.2 below, one should short the underlying asset when $S_t$ is far from $M_0^t$, and long this asset when $S_t$ becomes closer to $M_0^t$.




(a) Lookback put prices for fixed $M_0^t$. (b) Lookback put prices for fixed $S_t$.

Fig. 12.3: Graph of lookback put option prices (2D).

* The animation works in Acrobat Reader on the entire pdf file.

## 12.2 PDE Method

Since the couple $(S_t, M_0^t)$ is a Markov process, the price of the lookback put option at time $t \in [0,T]$ can be written as a function

$$\begin{aligned} f(t,S_t,M_0^t) &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[\phi\big(S_T,M_0^T\big)\,\big|\,\mathcal{F}_t\big] \\ &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[\phi\big(S_T,M_0^T\big)\,\big|\,S_t,\ M_0^t\big] \end{aligned} \tag{12.1}$$

of $S_t$ and $M_0^t$, $0 \leqslant t \leqslant T$.

**Black–Scholes PDE for lookback put option prices**

In Proposition 12.2 we derive the partial differential equation (PDE) for the pricing function $f(t,x,y)$ of a self-financing portfolio hedging a lookback put option. See Exercise 12.6 for the verification of the boundary conditions (12.3a)-(12.3c).

**Proposition 12.2.** *The function $f(t,x,y)$ defined by*

$$f(t,x,y) = \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[M_0^T - S_T\,\big|\,S_t = x,\ M_0^T = y\big], \quad t \in [0,T],\ x,y>0,$$

*is $\mathcal{C}^2((0,T)\times(0,\infty)^2)$ and satisfies the Black–Scholes PDE*

$$rf(t,x,y) = \frac{\partial f}{\partial t}(t,x,y) + rx\frac{\partial f}{\partial x}(t,x,y) + \frac{1}{2}x^2\sigma^2\frac{\partial^2 f}{\partial x^2}(t,x,y), \tag{12.2}$$

$0 \leqslant t \leqslant T$, $x,y>0$, *subject to the boundary conditions*

$$\begin{cases} f(t,0^+,y) = \mathrm{e}^{-(T-t)r}y, & 0 \leqslant t \leqslant T, \quad y \geqslant 0, & \text{(12.3a)} \\ \dfrac{\partial f}{\partial y}(t,x,y)_{|y=x} = 0, & 0 \leqslant t \leqslant T, \quad y>0, & \text{(12.3b)} \\ f(T,x,y) = y-x, & 0 \leqslant x \leqslant y. & \text{(12.3c)} \end{cases}$$

*The replicating portfolio of the lookback put option is given by*

$$\xi_t = \frac{\partial f}{\partial x}\big(t,S_t,M_0^t\big), \qquad t \in [0,T]. \tag{12.4}$$

*Proof.* The existence of $f(t,x,y)$ follows from the Markov property, more precisely, from the time homogeneity of the asset price process $(S_t)_{t\in\mathbb{R}_+}$ the function $f(t,x,y)$ satisfies

$$\begin{aligned} f(t,x,y) &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[\phi\big(S_T,M_0^T\big)\,\big|\,S_t=x,\ M_0^t=y\big] \\ &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[\phi\left(x\frac{S_T}{S_t},\mathrm{Max}\left(y,M_t^T\right)\right)\right] \end{aligned}$$



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$$= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[\phi\left(x\frac{S_{T-t}}{S_0}, \mathrm{Max}\left(y, M_0^{T-t}\right)\right)\right], \quad t \in [0,T].$$

Applying the change of variable formula to the discounted portfolio value

$$\widetilde{f}(t,x,y) := \mathrm{e}^{-rt}f(t,x,y) = \mathrm{e}^{-rT}\mathbb{E}^*\left[\phi(S_T, M_0^T)\,\middle|\, S_t = x,\ M_0^t = y\right]$$

which is a martingale indexed by $t \in [0,T]$ under $\mathbb{P}^*$, we have

$$\begin{aligned}
d\widetilde{f}\big(t,S_t,M_0^t\big) &= -r\,\mathrm{e}^{-rt}f\big(t,S_t,M_0^t\big)dt + \mathrm{e}^{-rt}df\big(t,S_t,M_0^t\big)\\
&= -r\,\mathrm{e}^{-rt}f\big(t,S_t,M_0^t\big)dt + \mathrm{e}^{-rt}\frac{\partial f}{\partial t}\big(t,S_t,M_0^t\big)dt + r\,\mathrm{e}^{-rt}S_t\frac{\partial f}{\partial x}\big(t,S_t,M_0^t\big)dt\\
&\quad + \mathrm{e}^{-rt}\sigma S_t\frac{\partial f}{\partial x}\big(t,S_t,M_0^t\big)dB_t + \frac{1}{2}\,\mathrm{e}^{-rt}\sigma^2S_t^2\frac{\partial^2 f}{\partial x^2}\big(t,S_t,M_0^t\big)dt\\
&\quad + \mathrm{e}^{-rt}\frac{\partial f}{\partial y}\big(t,S_t,M_0^t\big)dM_0^t, \qquad (12.5)
\end{aligned}$$

according to the following extension of the Itô multiplication table 12.1.

| $\cdot$ | $dt$ | $dB_t$ | $dM_0^t$ |
|---|---|---|---|
| $dt$ | 0 | 0 | 0 |
| $dB_t$ | 0 | $dt$ | 0 |
| $dM_0^t$ | 0 | 0 | 0 |

Table 12.1: Extended Itô multiplication table.

Since $\big(\widetilde{f}(t,S_t,M_0^t)\big)_{t\in[0,T]} = \big(\mathrm{e}^{-rT}\mathbb{E}^*\big[\phi(S_T,M_0^T)\,\big|\,\mathcal{F}_t\big]\big)_{t\in[0,T]}$ is a martingale under $\mathbb{P}$ and $\big(M_0^t\big)_{t\in[0,T]}$ has finite variation (it is in fact a non-decreasing process), (12.5) yields:

$$d\widetilde{f}\big(t,S_t,M_0^t\big) = \sigma S_t\frac{\partial \widetilde{f}}{\partial x}\big(t,S_t,M_0^t\big)dB_t, \qquad t \in [0,T], \qquad (12.6)$$

and the function $f(t,x,y)$ satisfies the equation

$$\begin{aligned}
&\frac{\partial f}{\partial t}\big(t,S_t,M_0^t\big)dt + rS_t\frac{\partial f}{\partial x}f\big(t,S_t,M_0^t\big)dt\\
&\qquad + \frac{1}{2}\sigma^2S_t^2\frac{\partial^2 f}{\partial x^2}\big(t,S_t,M_0^t\big)dt + \frac{\partial f}{\partial y}\big(t,S_t,M_0^t\big)dM_0^t = rf\big(t,S_t,M_0^t\big)dt, \qquad (12.7)
\end{aligned}$$

which implies

$$\frac{\partial f}{\partial t}\big(t,S_t,M_0^t\big) + rS_t\frac{\partial f}{\partial x}\big(t,S_t,M_0^t\big) + \frac{1}{2}\sigma^2S_t^2\frac{\partial^2 f}{\partial x^2}\big(t,S_t,M_0^t\big) = rf\big(t,S_t,M_0^t\big),$$

which is (12.2), and

$$\frac{\partial f}{\partial y}(t, S_t, M_0^t) dM_0^t = 0.$$

Indeed, $M_0^t$ increases only on a set of zero Lebesgue measure (which has no isolated points), therefore the Lebesgue measure $dt$ and the measure $dM_0^t$ are mutually *singular*, hence by the Lebesgue decomposition theorem, both components in $dt$ and $dM_0^t$ should vanish in (12.7) if the sum vanishes, see also the Cantor function. This implies

$$\frac{\partial f}{\partial y}(t, S_t, M_0^t) = 0,$$

when $dM_0^t > 0$, hence since

$$\{S_t = M_0^t\} \quad \Longleftrightarrow dM_0^t > 0$$

and

$$\{S_t < M_0^t\} \quad \Longleftrightarrow dM_0^t = 0,$$

we have

$$\frac{\partial f}{\partial y}(t, S_t, S_t) = \frac{\partial f}{\partial y}(t, x, y)_{x=S_t,\ y=S_t} = 0,$$

since $M_0^t$ hits $S_t$, *i.e.* $M_0^t = S_t$, only when $M_0^t$ increases at time $t$, and this shows the boundary condition (12.3b).

On the other hand, (12.6) shows that

$$\phi(S_T, M_0^T) = \mathbb{E}^*[\phi(S_T, M_0^T)] + \sigma \int_0^T S_t \frac{\partial f}{\partial x}(t, x, M_0^t)_{|x=S_t} dB_t,$$

$0 \leqslant t \leqslant T$, which implies (12.4) as in the proof of Propositions 6.1 or 11.3. □

In other words, the price of the lookback put option takes the form

$$f(t, S_t, M_0^t) = \mathrm{e}^{-(T-t)r} \mathbb{E}^*\big[M_0^T - S_T \,\big|\, \mathcal{F}_t\big],$$

where the function $f(t, x, y)$ is given from Proposition 12.1 as

$$\begin{aligned} f(t,x,y) = {} & y\,\mathrm{e}^{-(T-t)r}\Phi\big(-\delta_-^{T-t}(x/y)\big) + x\left(1+\frac{\sigma^2}{2r}\right)\Phi\big(\delta_+^{T-t}(x/y)\big) \\ & -x\frac{\sigma^2}{2r}\,\mathrm{e}^{-(T-t)r}\left(\frac{y}{x}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^{T-t}(y/x)\right) - x. \end{aligned} \tag{12.8}$$

**Remark 12.3.** *We have*



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$$f(t,x,x) = xC(T-t),$$

*with*

$$\begin{aligned} C(\tau) &= \mathrm{e}^{-r\tau}\Phi\left(-\delta_-^\tau(1)\right) + \left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\delta_+^\tau(1)\right) - \frac{\sigma^2}{2r}\mathrm{e}^{-r\tau}\Phi\left(-\delta_-^\tau(1)\right) - 1 \\ &= \mathrm{e}^{-r\tau}\Phi\left(-\frac{r-\sigma^2/2}{\sigma}\sqrt{\tau}\right) + \left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\frac{r+\sigma^2/2}{\sigma}\sqrt{\tau}\right) \\ &\quad -\frac{\sigma^2}{2r}\mathrm{e}^{-r\tau}\Phi\left(-\frac{r-\sigma^2/2}{\sigma}\sqrt{\tau}\right) - 1, \qquad \tau > 0, \end{aligned}$$

*hence*

$$\frac{\partial f}{\partial x}(t,x,x) = C(T-t), \qquad t \in [0,T].$$

**Scaling property of lookback put option prices**

From (12.8) and the following argument we note the scaling property

$$\begin{aligned} f(t,x,y) &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[M_0^T - S_T \,\middle|\, S_t = x,\ M_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[\mathrm{Max}\left(M_0^t, M_t^T\right) - S_T \,\middle|\, S_t = x,\ M_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[\mathrm{Max}\left(\frac{M_0^t}{S_t}, \frac{M_t^T}{S_t}\right) - \frac{S_T}{S_t}\,\middle|\, S_t = x,\ M_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[\mathrm{Max}\left(\frac{y}{x}, \frac{M_t^T}{x}\right) - \frac{S_T}{x}\,\middle|\, S_t = x,\ M_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[\mathrm{Max}\left(M_0^t, M_t^T\right) - S_T \,\middle|\, S_t = 1,\ M_0^t = \frac{y}{x}\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[M_0^T - S_T \,\middle|\, S_t = 1,\ M_0^t = \frac{y}{x}\right] \\ &= xf(t,1,y/x) \\ &= xg(T-t,x/y), \end{aligned}$$

where we let

$$g(\tau,z) :=$$

$$\frac{1}{z}\mathrm{e}^{-r\tau}\Phi\left(-\delta_-^\tau(z)\right) + \left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\delta_+^\tau(z)\right) - \frac{\sigma^2}{2r}\mathrm{e}^{-r\tau}\left(\frac{1}{z}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^\tau\left(\frac{1}{z}\right)\right) - 1,$$

with the boundary condition

$$\begin{cases} \dfrac{\partial g}{\partial z}(\tau, 1) = 0, & \tau > 0, \qquad (12.9a) \\ g(0, z) = \dfrac{1}{z} - 1, & z \in (0, 1]. \qquad (12.9b) \end{cases}$$

Figure 12.4 shows a graph of the function $g(\tau, z)$.

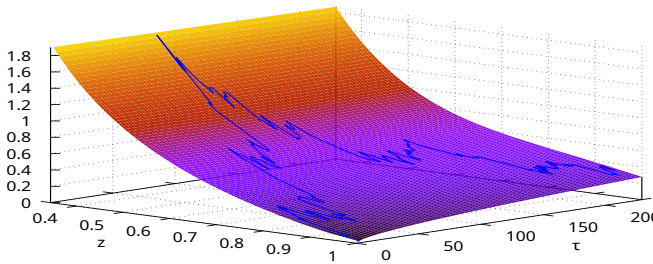



Fig. 12.4: Graph of the normalized lookback put option price.

Figure 12.5 represents the path of the underlying asset price used in Figures 12.4 to 12.7.

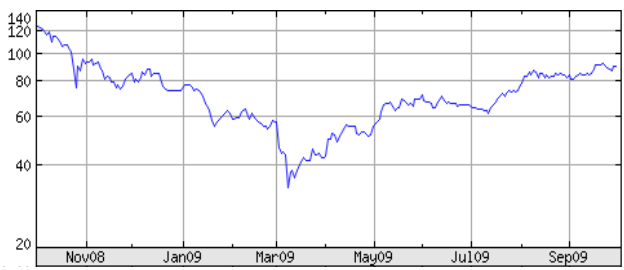



Fig. 12.5: Graph of underlying asset prices.

Figure 12.13 represents the corresponding underlying asset price and its running maximum.



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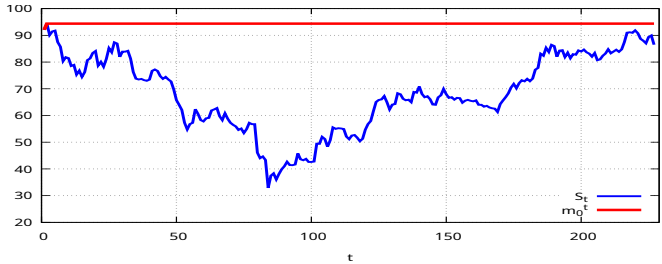



Fig. 12.6: Running maximum of the underlying asset price.

Next, we represent the option price as a function of time, together with the process $\left(M_0^t - S_t\right)_{t\in\mathbb{R}_+}$.

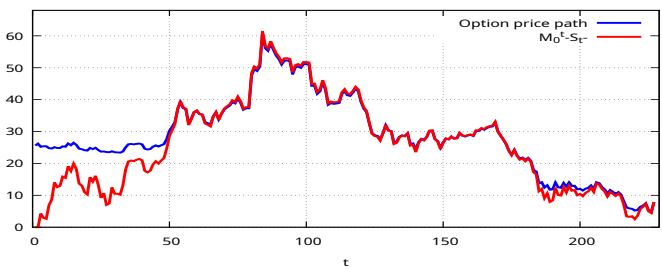



Fig. 12.7: Graph of the lookback put option price.

**Black–Scholes approximation of lookback put option prices**

Letting

$$\mathrm{Bl}_{\mathrm{p}}(x,K,r,\sigma,\tau) := K\,\mathrm{e}^{-r\tau}\Phi\left(-\delta_-^\tau\left(\frac{x}{K}\right)\right) - x\Phi\left(-\delta_+^\tau\left(\frac{x}{K}\right)\right)$$

denote the standard Black–Scholes formula for the price of the European put option.

**Proposition 12.4.** *The lookback put option price can be rewritten as*

$$\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[M_0^T - S_T \,\middle|\, \mathcal{F}_t\right] = \mathrm{Bl}_{\mathrm{p}}(S_t, M_0^t, r, \sigma, T-t) \tag{12.10}$$
$$+S_t\frac{\sigma^2}{2r}\left(\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) - \mathrm{e}^{-(T-t)r}\left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right)\right).$$

In other words, we have

$$\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[M_0^T - S_T \,\middle|\, \mathcal{F}_t\right] = \mathrm{Bl}_{\mathrm{p}}(S_t, M_0^t, r, \sigma, T-t) + S_t h_p\left(T-t, \frac{S_t}{M_0^t}\right)$$

where the function

$$h_p(\tau, z) = \frac{\sigma^2}{2r}\Phi\left(\delta_+^\tau(z)\right) - \frac{\sigma^2}{2r}\,\mathrm{e}^{-r\tau} z^{-2r/\sigma^2}\Phi\left(-\delta_-^\tau\left(\frac{1}{z}\right)\right), \tag{12.11}$$

depends only on time $\tau$ and $z = S_t/M_0^t$. In other words, due to the relation

$$\begin{aligned}\mathrm{Bl}_{\mathrm{p}}(x, y, r, \sigma, \tau) &= y\,\mathrm{e}^{-r\tau}\Phi\left(-\delta_-^\tau\left(\frac{x}{y}\right)\right) - x\Phi\left(-\delta_+^\tau\left(\frac{x}{y}\right)\right)\\ &= x\mathrm{Bl}_{\mathrm{p}}(1, y/x, r, \sigma, \tau)\end{aligned}$$

for the standard Black–Scholes put option pricing formula, we observe that $f(t, x, y)$ satisfies

$$f(t, x, y) = x\mathrm{Bl}_{\mathrm{p}}\left(1, \frac{y}{x}, r, \sigma, T-t\right) + xh\left(T-t, \frac{x}{y}\right),$$

*i.e.*

$$f(t, x, y) = xg\left(T-t, \frac{x}{y}\right),$$

with

$$g(\tau, z) = \mathrm{Bl}_{\mathrm{p}}\left(1, \frac{1}{z}, r, \sigma, \tau\right) + h_p(\tau, z), \tag{12.12}$$

where the function $h_p(\tau, z)$ is a correction term given by (12.11) which is small when $z = x/y$ or $\tau$ become small.

Note that $(x, y) \longmapsto xh_p(T-t, x/y)$ also satisfies the Black–Scholes PDE (12.2), in particular $(\tau, z) \longmapsto \mathrm{Bl}_{\mathrm{p}}(1, 1/z, r, \sigma, \tau)$ and $h_p(\tau, z)$ both satisfy the PDE

$$\frac{\partial h_p}{\partial \tau}(\tau, z) = z\left(r + \sigma^2\right)\frac{\partial h_p}{\partial z}(\tau, z) + \frac{1}{2}\sigma^2 z^2 \frac{\partial^2 h_p}{\partial z^2}(\tau, z), \tag{12.13}$$

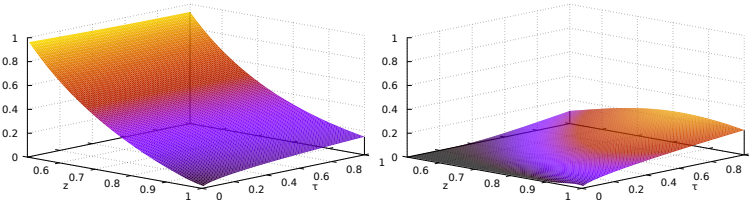
$\tau \in [0, T]$, $z \in [0, 1]$, subject to the boundary condition

$$h_p(0, z) = 0, \qquad 0 \leqslant z \leqslant 1.$$

Figure 12.8b illustrates the decomposition (12.12) of the normalized look-back put option price $g(\tau, z)$ in Figure 12.4 into the Black–Scholes put price function $\mathrm{Bl}_{\mathrm{p}}(1, 1/z, r, \sigma, \tau)$ and $h_p(\tau, z)$.



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(a) Lookback put price. (b) Correction term $h_p(\tau, z)$.

Fig. 12.8: Normalized Black–Scholes put price and correction term in (12.12).

Note that in Figure 12.8b the condition $h_p(0, z) = 0$ is not fully respected as $z$ tends to 1, due to numerical instabilities in the approximation of the function $\Phi$.

## 12.3 The Lookback Call Option

The standard Lookback call option gives the right to buy the underlying asset at its historically lowest price. In this case, the floating strike price is $m_0^T$ and the payoff is

$$C = S_T - m_0^T.$$

The following result gives the price of the lookback call option, see Eq. (11) in Goldman et al. (1979) and Proposition 9.5.1, page 270 of Dana and Jeanblanc (2007).

**Proposition 12.5.** *The price at time $t \in [0, T]$ of the lookback call option with payoff $S_T - m_0^T$ is given by*

$$\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[S_T - m_0^T \,\middle|\, \mathcal{F}_t\right] \\
&= S_t\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) - m_0^t\,\mathrm{e}^{-(T-t)r}\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) \\
&\quad + \mathrm{e}^{-(T-t)r}S_t\frac{\sigma^2}{2r}\left(\frac{m_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right) - S_t\frac{\sigma^2}{2r}\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right)
\end{aligned}$$

*Proof.* By Proposition 10.10 we have

$$\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[S_T - m_0^T \,\middle|\, \mathcal{F}_t\right] = S_t - \mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[m_0^T \,\middle|\, \mathcal{F}_t\right] \\
&= S_t\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) - \mathrm{e}^{-(T-t)r}m_0^t\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{m_0^t}\right)\right)
\end{aligned}$$

$$+\mathrm{e}^{-(T-t)r}\frac{S_t\sigma^2}{2r}\left(\left(\frac{m_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right)-\mathrm{e}^{(T-t)r}\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right)\right).$$

□

Figure 12.9 represents the price of the lookback call option as a function of $m_0^t$ and $S_t$ for different values of the time to maturity $T-t$.

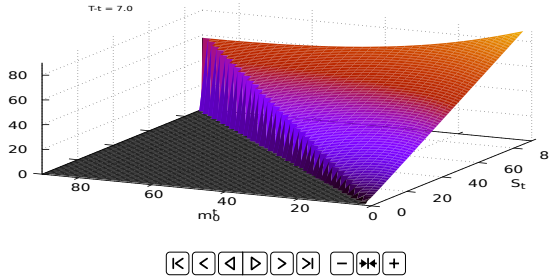
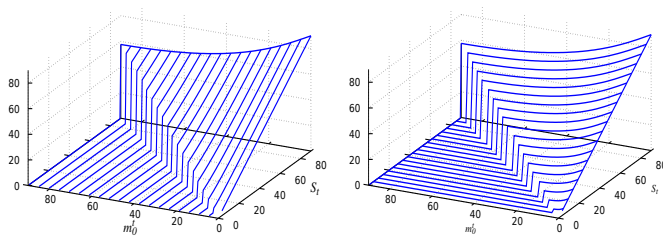



Fig. 12.9: Graph of the lookback call option price.*

From Figures 12.9 and 12.10, we note the following.

i) When the underlying asset price $S_t$ is far from $m_0^t$, an increase in the value $S_t$ clearly results into a higher call option price.
ii) When the underlying asset price $S_t$ is close to $m_0^t$, a decrease in $S_t$ could lead to a decrease in the value of $m_0^t$, however on average this appears insufficient to increase the average option payoff.




(a) Call prices as functions of $S$. (b) Call prices as functions of $m$.

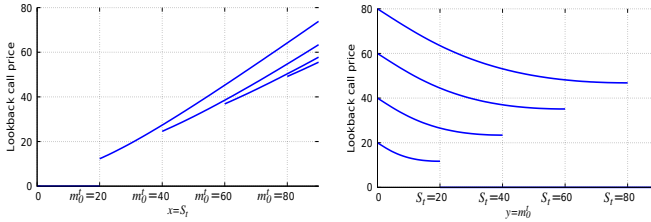
Fig. 12.10: Graph of lookback call option prices.

* The animation works in Acrobat Reader on the entire pdf file.



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Figures 12.10 and 12.11 show accordingly that, from the Delta hedging strategy for lookback call options, see Propositions 12.6 and 12.8, one should long the underlying asset in order to hedge a lookback call option.




(a) Lookback call prices for fixed $m_0^t$. (b) Lookback call prices for fixed $S_t$.

Fig. 12.11: Graphs of lookback call option prices (2D).

**Black–Scholes PDE for lookback call option prices**

Since the couple $(S_t, m_0^t)$ is also a Markov process, the price of the lookback call option at time $t \in [0,T]$ can be written as a function

$$\begin{aligned} f(t,S_t,m_0^t) &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[\phi(S_T,m_0^T)\,\big|\,\mathcal{F}_t\big] \\ &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[\phi(S_T,m_0^T)\,\big|\,S_t,\ m_0^t\big] \end{aligned}$$

of $S_t$ and $m_0^t$, $0 \leqslant t \leqslant T$. By the same argument as in the proof of Proposition 12.2, we obtain the following result.

**Proposition 12.6.** *The function $f(t,x,y)$ defined by*

$$f(t,x,y) = \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[S_T - m_0^T \,\big|\, S_t = x,\ m_0^t = y\big], \quad t \in [0,T],\ x,y>0,$$

*is $\mathcal{C}^2((0,T)\times(0,\infty)^2)$ and satisfies the Black–Scholes PDE*

$$rf(t,x,y) = \frac{\partial f}{\partial t}(t,x,y) + rx\frac{\partial f}{\partial x}(t,x,y) + \frac{1}{2}x^2\sigma^2\frac{\partial^2 f}{\partial x^2}(t,x,y),$$

*$0 \leqslant t \leqslant T$, $x>0$, subject to the boundary conditions*

$$\begin{cases} \lim\limits_{y\searrow 0} f(t,x,y) = x, & 0 \leqslant t \leqslant T, \quad x>0, \qquad (12.14a) \\ \dfrac{\partial f}{\partial y}(t,x,y)_{y=x} = 0, & 0 \leqslant t \leqslant T, \quad y>0, \qquad (12.14b) \\ f(T,x,y) = x-y, & 0 < y \leqslant x, \qquad (12.14c) \end{cases}$$



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*and the corresponding self-financing hedging strategy is given by*

$$\xi_t = \frac{\partial f}{\partial x}(t, S_t, m_0^t), \qquad t \in [0,T], \tag{12.15}$$

*which represents the quantity of the risky asset $S_t$ to be held at time $t$ in the hedging portfolio.*

In other words, the price of the lookback call option takes the form

$$f(t, S_t, m_t) = \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[S_T - m_0^T \,\big|\, \mathcal{F}_t\big],$$

where the function $f(t,x,y)$ is given by

$$\begin{aligned} f(t,x,y) &= x\Phi\left(\delta_+^{T-t}\left(\frac{x}{y}\right)\right) - \mathrm{e}^{-(T-t)r} y\Phi\left(\delta_-^{T-t}\left(\frac{x}{y}\right)\right) \\ &\quad + \mathrm{e}^{-(T-t)r} x\frac{\sigma^2}{2r}\left(\left(\frac{y}{x}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{y}{x}\right)\right) - \mathrm{e}^{(T-t)r}\Phi\left(-\delta_+^{T-t}\left(\frac{x}{y}\right)\right)\right) \\ &= x - y\,\mathrm{e}^{-(T-t)r}\Phi\left(\delta_-^{T-t}\left(\frac{x}{y}\right)\right) - x\left(1+\frac{\sigma^2}{2r}\right)\Phi\left(-\delta_+^{T-t}\left(\frac{x}{y}\right)\right) \\ &\quad + x\,\mathrm{e}^{-(T-t)r}\frac{\sigma^2}{2r}\left(\frac{y}{x}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{y}{x}\right)\right). \end{aligned} \tag{12.16}$$

**Scaling property of lookback call option prices**

We note the scaling property

$$\begin{aligned} f(t,x,y) &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[S_T - m_0^T \,\big|\, S_t = x,\ m_0^t = y\big] \\ &= \mathrm{e}^{-(T-t)r}\mathbb{E}^*\big[S_T - \min\left(m_0^t, m_t^T\right) \,\big|\, S_t = x,\ m_0^t = y\big] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[\frac{S_T}{S_t} - \min\left(\frac{m_0^t}{S_t}, \frac{m_t^T}{S_t}\right) \,\middle|\, S_t = x,\ m_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[\frac{S_T}{x} - \min\left(\frac{y}{x}, \frac{m_t^T}{x}\right) \,\middle|\, S_t = x,\ m_0^t = y\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[S_T - \min\left(m_0^t, m_t^T\right) \,\middle|\, S_t = 1,\ m_0^t = \frac{y}{x}\right] \\ &= \mathrm{e}^{-(T-t)r}x\mathbb{E}^*\left[S_T - m_0^T \,\middle|\, S_t = 1,\ m_0^t = \frac{y}{x}\right] \\ &= xf(t,1,y/x) \\ &= xg\left(T-t, \frac{1}{z}\right), \end{aligned}$$

where

$$g(\tau,z) := 1-\frac{1}{z}\,\mathrm{e}^{-r\tau}\Phi\left(\delta_-^\tau(z)\right)-\left(1+\frac{\sigma^2}{2r}\right)\Phi\left(-\delta_+^\tau(z)\right)+\frac{\sigma^2}{2r}\,\mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right),$$

with $g(\tau,1)=C(T-t)$, and

$$f(t,x,y)=xg\left(T-t,\frac{x}{y}\right)$$

and the boundary condition

$$\begin{cases}\dfrac{\partial g}{\partial z}(\tau,1)=0, & \tau>0, \qquad (12.17a)\\ g(0,z)=1-\dfrac{1}{z}, & z\geqslant 1. \qquad (12.17b)\end{cases}$$

Figure 12.12 shows a graph of the function $g(\tau,z)$.

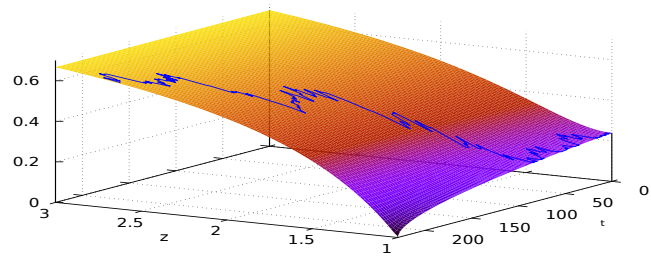



Fig. 12.12: Normalized lookback call option price.

Figures 12.12 to 12.14 use the path of underlying asset price displayed in Figure 12.5. Figure 12.13 represents the corresponding underlying asset price and its running minimum.

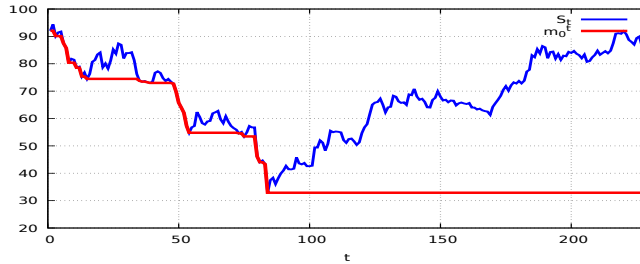



Fig. 12.13: Running minimum of the underlying asset price.

Next, we represent the option price as a function of time, together with the process $\left(S_t - m_0^t\right)_{t\in\mathbb{R}_+}$.

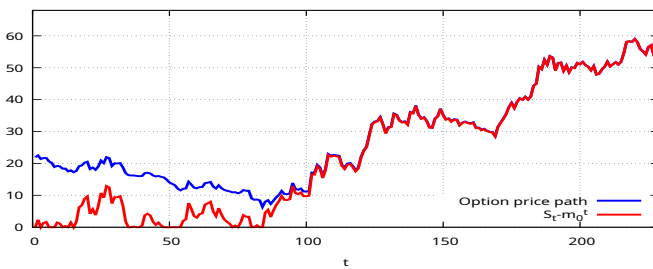



Fig. 12.14: Graph of the lookback call option price.

**Black–Scholes approximation of lookback call option prices**

Let

$$\mathrm{Bl_c}(S,K,r,\sigma,\tau) = S\Phi\left(\delta_+^\tau\left(\frac{S}{K}\right)\right) - K\,\mathrm{e}^{-r\tau}\Phi\left(\delta_-^\tau\left(\frac{S}{K}\right)\right)$$

denote the standard Black–Scholes formula for the price of the European call option.

**Proposition 12.7.** *The lookback call option price can be rewritten as*

$$\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[S_T - m_0^T \,|\, \mathcal{F}_t\right] = \mathrm{Bl_c}(S_t, m_0^t, r, \sigma, T-t) \tag{12.18}$$
$$-S_t\frac{\sigma^2}{2r}\left(\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) - \mathrm{e}^{-(T-t)r}\left(\frac{m_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right)\right).$$

In other words, we have

$$\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[S_T - m_0^T \,\middle|\, \mathcal{F}_t\right] := \mathrm{Bl}_{\mathrm{c}}(S_t, m_0^t, r, \sigma, T-t) + S_t h_c\left(T-t, \frac{S_t}{m_0^t}\right)$$

where the correction term

$$h_c(\tau, z) = -\frac{\sigma^2}{2r}\left(\Phi\left(-\delta_+^\tau(z)\right) - \mathrm{e}^{-r\tau} z^{-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\right), \tag{12.19}$$

is small when $z = S_t/m_0^t$ becomes large or $\tau$ becomes small. In addition, $h_p(\tau, z)$ is linked to $h_c(\tau, z)$ by the relation

$$h_c(\tau, z) = h_p(\tau, z) - \frac{\sigma^2}{2r}\left(1 - \mathrm{e}^{-r\tau} z^{-2r/\sigma^2}\right), \qquad \tau \geqslant 0, \quad z \geqslant 0,$$

where $(z, \tau) \longmapsto \mathrm{e}^{-r\tau} z^{-2r/\sigma^2}$ also solves the PDE (12.13). Due to the relation

$$\begin{aligned}\mathrm{Bl}_{\mathrm{c}}(x, y, r, \sigma, \tau) &= x\Phi\left(\delta_+^\tau\left(\frac{x}{y}\right)\right) - y\,\mathrm{e}^{-r\tau}\Phi\left(\delta_-^\tau\left(\frac{x}{y}\right)\right)\\ &= x\mathrm{Bl}_{\mathrm{c}}\left(1, \frac{y}{x}, r, \sigma, \tau\right)\end{aligned}$$

for the standard Black–Scholes call price formula, recall that from Proposition 12.7, $f(t, x, y)$ can be decomposed as

$$f(t, x, y) = x\mathrm{Bl}_{\mathrm{c}}\left(1, \frac{y}{x}, r, \sigma, T-t\right) + x h_c\left(T-t, \frac{x}{y}\right),$$

where $h_c(\tau, z)$ is the function given by (12.19), *i.e.*

$$f(t, x, y) = x g\left(T-t, \frac{x}{y}\right),$$

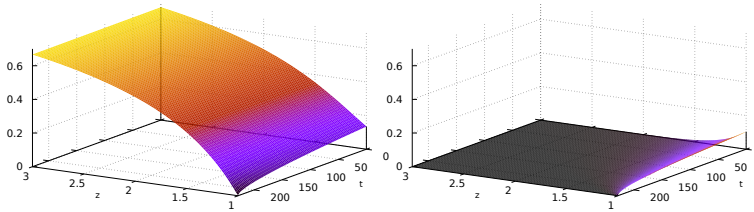
with

$$g(\tau, z) = \mathrm{Bl}_{\mathrm{c}}\left(1, \frac{1}{z}, r, \sigma, \tau\right) + h_c(\tau, z), \tag{12.20}$$

where $(x, y) \longmapsto x h_c(T-t, x/y)$ also satisfies the Black–Scholes PDE (12.2), *i.e.* $(\tau, z) \longmapsto \mathrm{Bl}_{\mathrm{c}}(1, 1/z, r, \sigma, \tau)$ and $h_c(\tau, z)$ both satisfy the PDE (12.13) subject to the boundary condition

$$h_c(0, z) = 0, \qquad z \geqslant 1.$$

Figures 12.15a and 12.15b show the decomposition of $g(t, z)$ in (12.20) and Figures 12.12-12.5 into the sum of the Black–Scholes call price function $\mathrm{Bl}_{\mathrm{c}}(1, 1/z, r, \sigma, \tau)$ and $h(t, z)$.




(a) Lookback call price $g(\tau, z)$. (b) Correction term $h_c(\tau, z)$.

Fig. 12.15: Normalized Black–Scholes call price and correction term in (12.20).

We also note that

$$\begin{aligned}
\mathbb{E}^*\big[M_0^T - m_0^T \,\big|\, S_0 = x\big] &= x - x\,\mathrm{e}^{-(T-t)r}\Phi\big(\delta_-^{T-t}(1)\big) \\
&\quad -x\left(1+\frac{\sigma^2}{2r}\right)\Phi\big(-\delta_+^{T-t}(1)\big) + x\,\mathrm{e}^{-(T-t)r}\frac{\sigma^2}{2r}\Phi\big(\delta_-^{T-t}(1)\big) \\
&\quad +x\,\mathrm{e}^{-(T-t)r}\Phi\big(-\delta_-^{T-t}(1)\big) + x\left(1+\frac{\sigma^2}{2r}\right)\Phi\big(\delta_+^{T-t}(1)\big) \\
&\quad -x\frac{\sigma^2}{2r}\,\mathrm{e}^{-(T-t)r}\Phi\big(-\delta_-^{T-t}(1)\big) - x \\
&= x\left(1+\frac{\sigma^2}{2r}\right)\Big(\Phi\big(\delta_+^{T-t}(1)\big) - \Phi\big(-\delta_+^{T-t}(1)\big)\Big) \\
&\quad +x\,\mathrm{e}^{-(T-t)r}\left(\frac{\sigma^2}{2r}-1\right)\Big(\Phi\big(\delta_-^{T-t}(1)\big) - \Phi\big(-\delta_-^{T-t}(1)\big)\Big).
\end{aligned}$$

## 12.4 Delta Hedging for Lookback Options

In this section we compute hedging strategies for lookback call and put options by application of the Delta hedging formula (12.15). See Bermin (1998), § 2.6.1, page 29, for another approach to the following result, using the Clark–Ocone formula. Here we use (12.15) instead, cf. Proposition 4.6 of El Khatib and Privault (2003).

**Proposition 12.8.** *The Delta hedging strategy of the lookback call option is given by*

$$\begin{aligned}
\xi_t &= 1 - \left(1+\frac{\sigma^2}{2r}\right)\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) \\
&\quad + \mathrm{e}^{-(T-t)r}\left(\frac{m_0^t}{S_t}\right)^{2r/\sigma^2}\left(\frac{\sigma^2}{2r}-1\right)\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right), \quad 0 \leqslant t \leqslant T.
\end{aligned} \tag{12.21}$$

*Proof.* By (12.15) and (12.18), we need to differentiate

$$f(t,x,y) = \mathrm{Bl_c}(x,y,r,\sigma,T-t) + xh_c\left(T-t, \frac{x}{y}\right)$$

with respect to the variable $x$, where

$$h_c(\tau,z) = -\frac{\sigma^2}{2r}\left(\Phi\left(-\delta_+^\tau(z)\right) - \mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\right)$$

is given by (12.19) First, we note that the relation

$$\frac{\partial}{\partial x}\mathrm{Bl_c}(x,y,r,\sigma,\tau) = \Phi\left(\delta_+^\tau\left(\frac{x}{y}\right)\right)$$

is known, cf. Propositions 6.4 and 7.13. Next, we have

$$\frac{\partial}{\partial x}\left(xh_c\left(\tau,\frac{x}{y}\right)\right) = h_c\left(\tau,\frac{x}{y}\right) + \frac{x}{y}\frac{\partial h_c}{\partial z}\left(\tau,\frac{x}{y}\right),$$

and

$$\begin{aligned}
&\frac{\partial h_c}{\partial z}(\tau,z) = -\frac{\sigma^2}{2r}\left(\frac{\partial}{\partial z}\left(\Phi\left(-\delta_+^\tau(z)\right)\right) - \mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\frac{\partial}{\partial z}\left(\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\right)\right)\\
&-\frac{\sigma^2}{2r}\left(\frac{2r}{\sigma^2}\mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\right)\\
&= \frac{\sigma}{2rz\sqrt{2\pi\tau}}\exp\left(-\frac{1}{2}\left(\delta_+^\tau(z)\right)^2\right)\\
&-\mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\frac{\sigma}{2rz\sqrt{2\pi\tau}}\exp\left(-\frac{1}{2}\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)^2\right) - \frac{2r}{\sigma^2}\mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right).
\end{aligned}$$

Next, we note that

$$\begin{aligned}
&\mathrm{e}^{-\left(\delta_-^\tau(1/z)\right)^2/2} = \exp\left(-\frac{1}{2}\left(\delta_+^\tau(z)\right)^2 - \frac{1}{2}\left(\frac{4r^2}{\sigma^2}\tau - \frac{4r}{\sigma}\delta_+^\tau(z)\sqrt{\tau}\right)\right)\\
&= \mathrm{e}^{-\left(\delta_+^\tau(z)\right)^2/2}\exp\left(-\frac{1}{2}\left(\frac{4r^2}{\sigma^2}\tau - \frac{4r}{\sigma^2}\left(\log z + \left(r+\frac{1}{2}\sigma^2\right)\tau\right)\right)\right)\\
&= \mathrm{e}^{-\left(\delta_+^\tau(z)\right)^2/2}\exp\left(\frac{-2r^2}{\sigma^2}\tau + \frac{2r}{\sigma^2}\log z + \frac{2r^2}{\sigma^2}\tau + r\tau\right)\\
&= \mathrm{e}^{r\tau}z^{2r/\sigma^2}\,\mathrm{e}^{-\left(\delta_+^\tau(z)\right)^2/2}
\end{aligned} \tag{12.22}$$

as in the proof of Proposition 6.4, hence

$$\frac{\partial h_c}{\partial z}\left(\tau,\frac{x}{y}\right) = -\mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right),$$

and

$$\frac{\partial}{\partial x}\left(xh_c\left(\tau,\frac{x}{y}\right)\right)=h_c\left(\tau,\frac{x}{y}\right)-\mathrm{e}^{-r\tau}\left(\frac{y}{x}\right)^{2r/\sigma^2}\Phi\left(\delta_-^\tau\left(\frac{y}{x}\right)\right),$$

which concludes the proof. □

We note that $\xi_t = 1 > 0$ as $T$ tends to infinity, and that at maturity $t = T$, the delta hedging strategy satisfies

$$\xi_T=\begin{cases}1 & \text{if } m_0^T < S_T,\\ 1-\frac{1}{2}\left(1+\frac{\sigma^2}{2r}\right)+\frac{1}{2}\left(\frac{\sigma^2}{2r}-1\right)=0 & \text{if } m_0^T = S_T.\end{cases}$$

In Figure 12.16 we represent the Delta of the lookback call option, as given by (12.21).

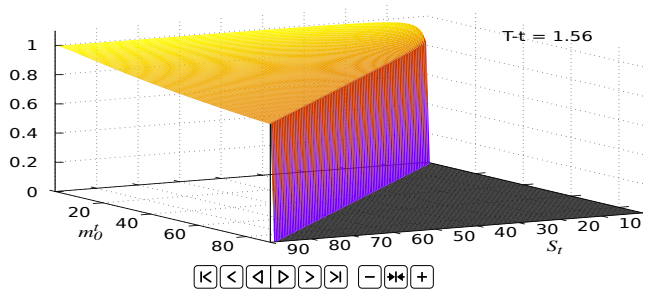



Fig. 12.16: Delta of the lookback call option with $r = 2\%$ and $\sigma = 0.41$.*

The above scaling procedure can be applied to the Delta of lookback call options by noting that $\xi_t$ can be written as

$$\xi_t=\zeta\left(t,\frac{S_t}{m_0^t}\right),$$

where the function $\zeta(t,z)$ is given by

$$\begin{aligned}\zeta(t,z)=\Phi\left(\delta_+^{T-t}(z)\right)-\frac{\sigma^2}{2r}\Phi\left(-\delta_+^{T-t}(z)\right)\\ +\mathrm{e}^{-(T-t)r}z^{-2r/\sigma^2}\left(\frac{\sigma^2}{2r}-1\right)\Phi\left(\delta_-^{T-t}\left(\frac{1}{z}\right)\right),\end{aligned}\tag{12.23}$$

$t \in [0,T]$, $z \in [0,1]$. The graph of the function $(t,z) \mapsto \zeta(t,z)$ is given in Figure 12.17.

* The animation works in Acrobat Reader on the entire pdf file.



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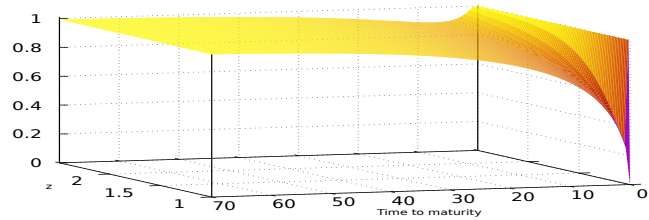



Fig. 12.17: Rescaled portfolio strategy for the lookback call option.

Similar calculations using (12.4) can be carried out for other types of lookback options, such as options on extrema and partial lookback options, cf. El Khatib (2003). As a consequence of Propositions 12.5 and 12.8, we have

$$
\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[S_T - m_0^T \,|\, \mathcal{F}_t\right] \\
&= S_t\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) - m_0^t\,\mathrm{e}^{-(T-t)r}\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) \\
&\quad + \mathrm{e}^{-(T-t)r}S_t\frac{\sigma^2}{2r}\left(\frac{m_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right) - S_t\frac{\sigma^2}{2r}\Phi\left(-\delta_+^{T-t}\left(\frac{S_t}{m_0^t}\right)\right) \\
&= \xi_t S_t + m_0^t\,\mathrm{e}^{-(T-t)r}\left(\left(\frac{S_t}{m_0^t}\right)^{1-2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{S_t}{m_0^t}\right)\right)\right),
\end{aligned}
$$

and the quantity of the riskless asset $\mathrm{e}^{rt}$ in the portfolio is given by

$$
\eta_t = m_0^t\,\mathrm{e}^{-rT}\left(\left(\frac{S_t}{m_0^t}\right)^{1-2r/\sigma^2}\Phi\left(\delta_-^{T-t}\left(\frac{m_0^t}{S_t}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{S_t}{m_0^t}\right)\right)\right),
$$

so that the portfolio value $V_t$ at time $t$ satisfies

$$
V_t = \xi_t S_t + \eta_t\,\mathrm{e}^{rt}, \qquad t \geqslant 0.
$$

**Proposition 12.9.** *The Delta hedging strategy of the lookback put option is given by*

$$
\begin{aligned}
\xi_t &= \left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) \qquad (12.24)\\
&\quad + \mathrm{e}^{-(T-t)r}\left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2}\left(1-\frac{\sigma^2}{2r}\right)\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right) - 1, \quad 0 \leqslant t \leqslant T.
\end{aligned}
$$

*Proof.* By (12.15) and (12.10), we need to differentiate

$$f(t,x,y) = \mathrm{Bl_p}(x,y,r,\sigma,T-t) + xh_p\left(T-t, \frac{x}{y}\right)$$

where

$$h_p(\tau,z) = \frac{\sigma^2}{2r}\Phi\left(\delta_+^\tau(z)\right) - \mathrm{e}^{-r\tau}\frac{\sigma^2}{2r}z^{-2r/\sigma^2}\Phi\big(-\delta_-^\tau(1/z)\big),$$

and

$$\delta_\pm^\tau(z) := \frac{1}{\sigma\sqrt{\tau}}\left(\log z + \left(r \pm \frac{1}{2}\sigma^2\right)\tau\right), \qquad z > 0.$$

We have

$$\begin{aligned}\frac{\partial h_p}{\partial z}(\tau,z) &= \frac{\sigma^2}{2r}\delta_+^{\prime\tau}(z)\,\varphi\left(\delta_+^\tau(z)\right) + \mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(-\delta_-^\tau\left(\frac{1}{z}\right)\right)\\ &\quad + \frac{\sigma^2}{2rz^2}\delta_-^{\prime\tau}\left(\frac{1}{z}\right)\mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\varphi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\\ &= \mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(-\delta_-^\tau\left(\frac{1}{z}\right)\right)\\ &\quad + \frac{\sigma}{2rz\sqrt{\tau}}\big(\varphi\big(\delta_+^\tau(z)\big) - \mathrm{e}^{-r\tau}z^{-2r/\sigma^2}\varphi\left(\delta_-^\tau\left(\frac{1}{z}\right)\right)\big).\end{aligned}$$

From the relation

$$\begin{aligned}\big(\delta_+^{T-t}(z)\big)^2 - \left(\delta_-^{T-t}\left(\frac{1}{z}\right)\right)^2 &= \left(\delta_+^{T-t}(z) + \delta_-^{T-t}\left(\frac{1}{z}\right)\right)\left(\delta_+^{T-t}(z) - \delta_-^{T-t}\left(\frac{1}{z}\right)\right)\\ &= \frac{2r}{\sigma^2}\log z + 2r(T-t),\end{aligned}$$

we have

$$\varphi\big(\delta_+^{T-t}(z)\big) = z^{-2r/\sigma^2}\,\mathrm{e}^{-r(T-t)}\varphi\left(\delta_-^{T-t}\left(\frac{1}{z}\right)\right),$$

hence

$$\frac{\partial h_p}{\partial z}(\tau,z) = \mathrm{e}^{-r\tau}z^{-1-2r/\sigma^2}\Phi\left(-\delta_-^\tau\left(\frac{1}{z}\right)\right).$$

Therefore, knowing that the Black–Scholes put Delta is

$$-\Phi\left(-\delta_+^{T-t}\left(\frac{x}{y}\right)\right) = -1 + \Phi\left(\delta_+^{T-t}\left(\frac{x}{y}\right)\right),$$

see *e.g.* Proposition 6.7, we have

$$\begin{aligned}\frac{\partial f}{\partial x}(t,x,y) &= -\Phi\left(-\delta_+^{T-t}\left(\frac{x}{y}\right)\right) + h_p\left(T-t,\frac{x}{y}\right) + \frac{x}{y}\frac{\partial h_p}{\partial z}\left(T-t,\frac{x}{y}\right)\\ &= -\Phi\left(-\delta_+^{T-t}\left(\frac{x}{y}\right)\right) + \frac{\sigma^2}{2r}\Phi\left(\delta_+^{T-t}\left(\frac{x}{y}\right)\right)\end{aligned}$$

$$+\mathrm{e}^{-(T-t)r}\left(\frac{y}{x}\right)^{2r/\sigma^2}\left(1-\frac{\sigma^2}{2r}\right)\Phi\left(-\delta_-^{T-t}\left(\frac{y}{x}\right)\right),$$

which yields (12.24). □

Note that we have $\xi_t = \sigma^2/(2r) > 0$ as $T$ tends to infinity. At maturity $t = T$, the delta hedging strategy satisfies

$$\xi_T = \begin{cases} -1 & \text{if } M_0^T > S_T, \\ \dfrac{1}{2}+\dfrac{\sigma^2}{4r}+\dfrac{1}{2}\left(1-\dfrac{\sigma^2}{2r}\right)-1=0 & \text{if } M_0^T = S_T. \end{cases}$$

In Figure 12.18 we represent the Delta of the lookback put option, as given by (12.24).

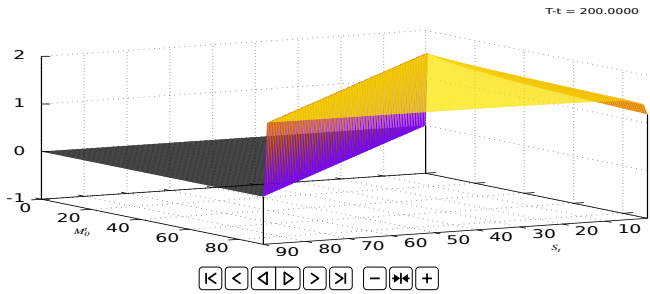



Fig. 12.18: Delta of the lookback put option with $r = 2\%$ and $\sigma = 0.25$.*

As a consequence of Propositions 12.1 and 12.9, we have

$$\begin{aligned}
&\mathrm{e}^{-(T-t)r}\mathbb{E}^*\left[M_0^T - S_T \,\middle|\, \mathcal{F}_t\right] \\
&= M_0^t\,\mathrm{e}^{-(T-t)r}\Phi\left(-\delta_-^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) + S_t\left(1+\frac{\sigma^2}{2r}\right)\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) \\
&\quad -S_t\,\mathrm{e}^{-(T-t)r}\frac{\sigma^2}{2r}\left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2}\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right) - S_t \\
&= \xi_t S_t + M_0^t\,\mathrm{e}^{-(T-t)r}\left(\Phi\left(-\delta_-^{T-t}\left(\frac{S_t}{M_0^t}\right)\right) - \left(\frac{S_t}{M_0^t}\right)^{1-2r/\sigma^2}\Phi\left(-\delta_-^{T-t}\left(\frac{M_0^t}{S_t}\right)\right)\right),
\end{aligned}$$

and the quantity of the riskless asset $\mathrm{e}^{rt}$ in the portfolio is given by

* The animation works in Acrobat Reader on the entire pdf file.

$$\eta_t = M_0^t \, \mathrm{e}^{-rT} \left( \Phi\left( -\delta_-^{T-t}\left( \frac{S_t}{M_0^t} \right) \right) - \left( \frac{S_t}{M_0^t} \right)^{1-2r/\sigma^2} \Phi\left( -\delta_-^{T-t}\left( \frac{M_0^t}{S_t} \right) \right) \right)$$

so that the portfolio value $V_t$ at time $t$ satisfies

$$V_t = \xi_t S_t + \eta_t \, \mathrm{e}^{rt}, \qquad t \geqslant 0.$$

## Exercises

Exercise 12.1

a) Give the probability density function of the maximum of drifted Brownian motion $\operatorname{Max}_{t\in[0,1]}(B_t + \sigma t/2)$.

b) Taking $S_t := \mathrm{e}^{\sigma B_t - \sigma^2 t/2}$, compute the expected value

$$\begin{aligned} \mathbb{E}\left[ \min_{t\in[0,1]} S_t \right] &= \mathbb{E}\left[ \min_{t\in[0,1]} \mathrm{e}^{\sigma B_t - \sigma^2 t/2} \right] \\ &= \mathbb{E}\left[ \mathrm{e}^{-\sigma \operatorname{Max}_{t\in[0,1]}(B_t + \sigma t/2)} \right]. \end{aligned}$$

c) Compute the "optimal exercise" price $\mathbb{E}\left[ \left( K - S_0 \min_{t\in[0,1]} \mathrm{e}^{\sigma B_t - \sigma^2 t/2} \right)^+ \right]$ of a finite expiration American put option with $S_0 \leqslant K$.

Exercise 12.2 Let $(B_t)_{t\in\mathbb{R}_+}$ denote a standard Brownian motion.

a) Compute the expected value

$$\mathbb{E}\left[ \operatorname*{Max}_{t\in[0,1]} S_t \right] = \mathbb{E}\left[ \mathrm{e}^{\sigma \operatorname{Max}_{t\in[0,1]}(B_t - \sigma t/2)} \right].$$

b) Compute the "optimal exercise" price $\mathbb{E}\left[ \left( S_0 \operatorname*{Max}_{t\in[0,1]} \mathrm{e}^{\sigma B_t - \sigma^2 t/2} - K \right)^+ \right]$ of a finite expiration American call option with $S_0 \geqslant K$.

Exercise 12.3 Consider a risky asset whose price $S_t$ is given by

$$dS_t = \sigma S_t dB_t + \frac{\sigma^2}{2} S_t dt,$$

where $(B_t)_{t\in\mathbb{R}_+}$ is a standard Brownian motion.

a) Compute the cumulative distribution function and the probability density function of the minimum $\min_{t\in[0,T]} B_t$ over the interval $[0,T]$?



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b) Compute the price value

$$\mathrm{e}^{-\sigma^2 T/2}\mathbb{E}^*\left[S_T - \min_{t\in[0,T]} S_t\right]$$

of a lookback call option on $S_T$ with maturity $T$.

Exercise 12.4

a) Compute the expected average $\mathbb{E}\left[\left(M_0^T + m_0^T\right)/2\right]$ of the maximum and minimum of *geometric Brownian motion* on the time interval $[0,T]$ without using the function $\Phi$, in both cases $r \neq 0$ and $r = 0$.
b) Find a call-put parity relation between lookback call and put options, in both cases $r \neq 0$ and $r = 0$.

Exercise 12.5 (Dassios and Lim (2019)) The digital drawdown call option with qualifying period pays a unit amount when the drawdown period reaches one unit of time, if this happens before fixed maturity $T$, but only if the size of drawdown at this stopping time is larger than a prespecified $K$. This provides an insurance against a prolonged drawdown, if the drawdown amount is large. Specifically, the digital drawdown call option is priced as

$$\mathbb{E}^*\left[\mathrm{e}^{-r\tau}\mathbb{1}_{\{\tau\leqslant T\}}\mathbb{1}_{\{M_0^\tau - S_\tau \geqslant K\}}\right],$$

where $M_0^t := \mathrm{Max}_{u\in[0,t]} S_u$, $U_t := t - \mathrm{Sup}\{0 \leqslant u \leqslant t \ : \ M_0^t = S_u\}$, and $\tau := \inf\{t \in \mathbb{R}_+ \ : \ U_t = 1\}$. Write the price of the drawdown option as a triple integral using the joint probability density function $f_{(\tau,S_\tau,M_\tau)}(t,x,y)$ of $(\tau, S_\tau, M_\tau)$ under the risk-neutral probability measure $\mathbb{P}^*$.

Exercise 12.6

a) Check explicitly that the boundary conditions (12.3a)-(12.3c) are satisfied.
b) Check explicitly that the boundary conditions (12.14a)-(12.14b) are satisfied.



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